



Mathematicians and programming

The pure, the tortured and the applied

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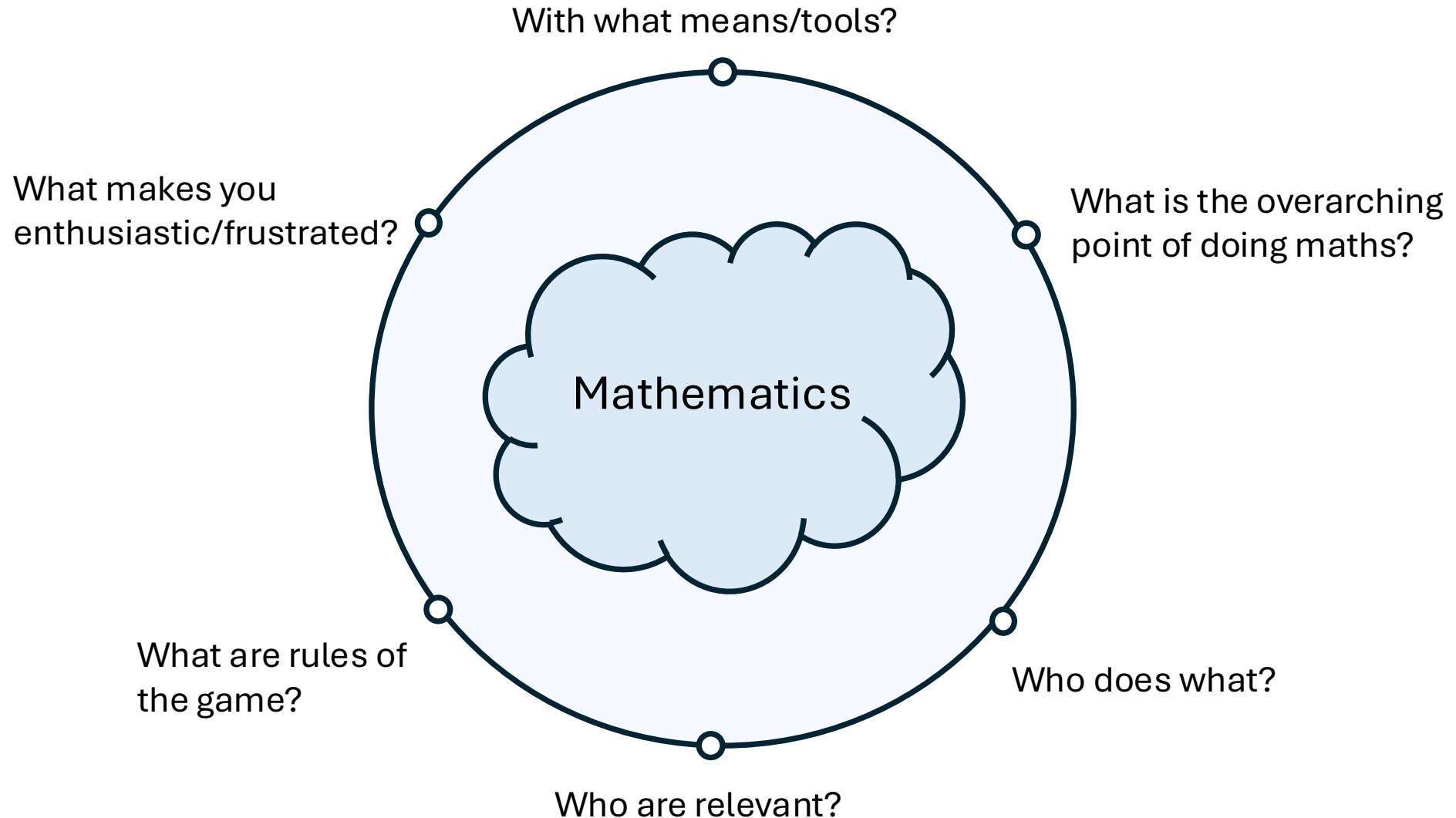
«Theorems» from educational research

- **Fundamental Theorem of Education** (Roth and Radford, 2011)
Students cannot know what they have yet to learn, and teachers cannot unlearn what they already know.
- **Hidden Curriculum Theorem** (Jackson, 1968; Engeström, 2015)
It's not just what we say, it's what we do.
- **Active Learning Theorem** (Freeman et al 2014)
Active learning beats passive learning.
- **Bloom's 2-Sigma Theorem** (Bloom, 1984; von Hippel, 2024)
When individually tutored by an expert, anyone can learn anything.
- **Vygotsky's Affective-Intellectual Theorem** (Vygotsky, 1978)
When you don't care, learning is superficial and brittle.
- **Motivation Stabilisation Theorem** (Engeström, 2015; Lave & Wenger, 1991)
When feeling isolated, motivation has a short half-life.

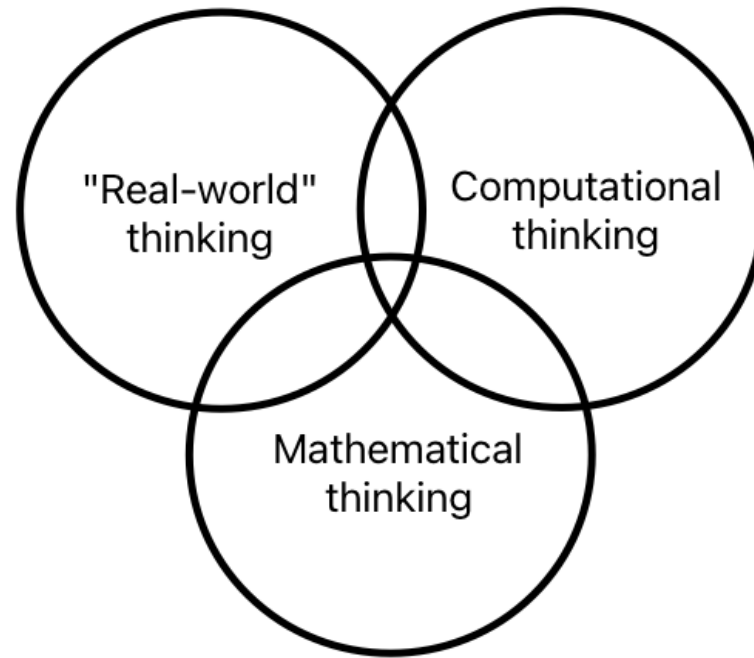
Quick questions

- How do you use computations/programming in your research?
- How do you use computations/programming in your courses?
- How does AI impact your research and teaching?

Mathematics is never just mathematics



Mathematicians' epistemic compass

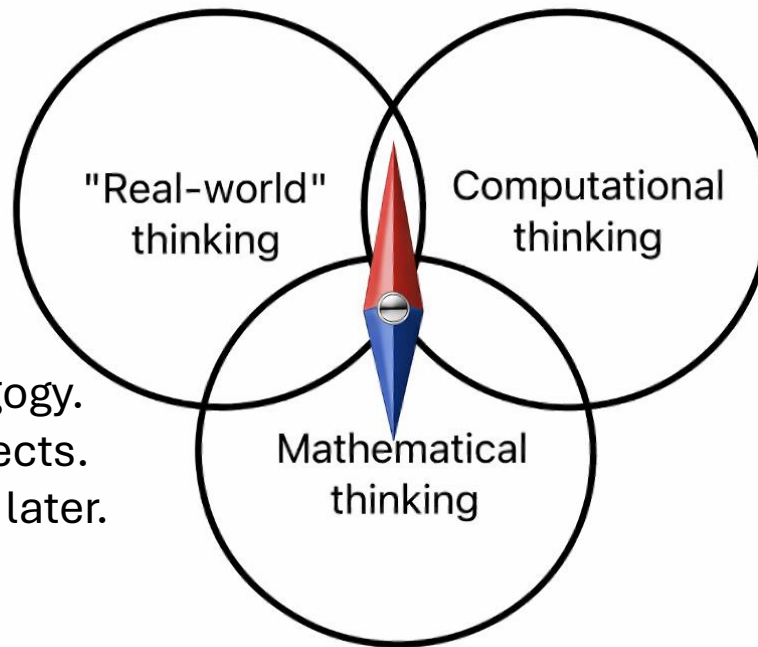


Mathematicians' epistemic compass

Applied math: Research must be directed toward application outside of maths to be of interest. Three variants: Methods focus, implementation focus, modelling focus.

Computations in teaching:

Methods focus: Simulations part of pedagogy.
Implementation focus: Programming projects.
Modelling focus: Theory first, programming later.



Numerical math: Development of theory for efficient algorithms to solve mathematical models. Simulations for "proof of concept" included in publications.

Computations in teaching:

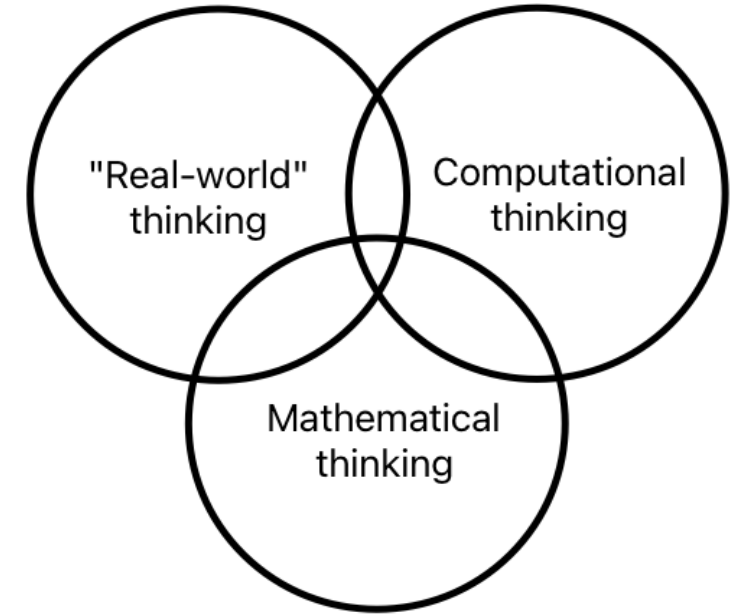
Implementation is not mathematics.
Theoretical v algorithmic insight.
Theory courses v methods courses.

Pure math: Theory as sole focus.
The truth, the whole truth, and nothing but the truth!

Computations in teaching:
Just cosmetic versus conceptual on-ramp.
Symbolic/exact versus numerical/approximative.
Distraction versus vocational necessity.

Example

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$$



Ways to leverage computations:

- To motivate definition of divergence.
- To illustrate definition of divergence.
- To illustrate issues of efficient of implementation.
- To get insight on real world phenomenon.
- To illustrate link between theoretical maths and real world phenomenon.

Questions for further discussion?